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## Seamless mode change between grid-connected mode and islanded mode for pv systems

### Abstract

A unified control scheme to achieve seamless mode change between grid-connected mode and islanded mode for photovoltaic (PV) systems includes two decoupled control units: a DC-DC converter control unit and a DC-AC converter control unit. The DC-bus voltage is controlled through the DC-DC converter control unit, which supplies a “stiff DC source” to the DC-AC converter. The seamless mode change is implemented in the DC-AC converter control module via three digital switches between power flow control for grid-connected mode (or MPPT mode), and droop control for islanded mode (or droop mode), while preserving the continuity of control signals in all control units with the configurations of digital switches.

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## Background/Summary

CROSS REFERENCE TO RELATED PATENT APPLICATIONS (1) This patent application claims the priority and benefit, under 35 U.S.C. § 119(e), of U.S. Provisional Patent Application Ser. No. 63/194,004, filed May 27, 2021, and titled “SEAMLESS MODE CHANGE BETWEEN GRID-CONNECTED MODE AND ISLANDED MODE FOR PV SYSTEMS”. U.S. Provisional Application Ser. No. 63/194,004 is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(1) Embodiments are generally related to the field of energy production. Embodiments are further

related to the field of photovoltaics. Embodiments are also related to methods, systems, and devices for grid-connected photovoltaic systems. Embodiments are related to methods, systems, and devices for islanded photovoltaic systems. Embodiments are further related to methods, systems, and devices for seamless mode change between grid-connected mode and islanded mode for photovoltaic systems.

BACKGROUND

(2) Photovoltaics (PV) are a promising solar technology that convert sunlight into electricity. PV enjoys broad applicability in the energy industry. Typical grid-integration PV systems use dual-stage power processing converters that include a DC-DC converter and a DC-AC converter, connected in series via a DC-voltage bus. In typical grid-integration PV systems, certain control aspects (e.g., power tracking algorithms) are integrated in the DC-DC converter control, while the DC-bus voltage is controlled through the design of the DC-AC converter control.

(3) For example, in some cases, a proportional-integral (PI) control is used for regulation of DC-bus voltage, to form a virtual current reference. The current control is adopted for AC grid integration of the DC-AC converter. This control framework works for PV systems in grid-connected mode.

(4) However, in addition to grid-connected mode, PV systems should also be designed to operate in islanded mode. For example, the PV source and the utility grid can work together to support some local loads, and the extra solar energy can be sent to the utility grid in grid-connected mode. If the utility grid is suddenly lost, the PV source still can support the local loads in islanded mode.

(5) The microgrid system will play a very important role in the future of the energy industry, and has received increasing attention in recent years. Specifically, it would be advantageous for PV systems to operate with microgrids and to incorporate functionality allowing both grid-connected mode and islanded mode.

(6) The operation of islanded mode is very important for distributed renewable generators, including solar power, particularly in microgrid applications. If the microgrids do not have energy storage devices (e.g., PV/diesel-generator microgrids, PV/fuel-cell microgrids, etc.), the energy storage devices are full, or the storage devices cannot store more renewable power, the distributed renewable generators need to operate in islanded mode to avoid the increase of both voltage and frequency.

(7) As a result, mode change between grid-connected mode and islanded mode is critical for grid-integration PV systems. Some research has focused on mode change of PV systems with a single-stage converter. However, in these studies the DC input voltage (the same as the PV output voltage) of the DC-AC converter cannot maintain a constant level, which affects its ability to reliably integrate with the electrical grid.

(8) As for grid-integration PV systems with dual-stage converters, it is a continuing challenge to design a system that properly changes operational modes because, in prior art control systems, the tracking is handled by the DC-DC converter control in grid-connected operation, and requires that the same amount of real power generated by PV sources be fed into the grid through the DC-AC converter control, in order to guarantee system stability. In this way, the DC-DC converter control and DC-AC converter control are coupled with each other. Therefore, the DC-AC converter cannot be dispatched with droop control in islanded mode directly.

(9) It would be useful to have an extra configuration to facilitate mode change where the whole control system should be switched between two different control configurations. However, solving this problem has proved to be a challenge. Specifically, attempts to address this issue have resulted in extreme complexity in both DC-DC converter control and DC-AC converter control, along with complex procedures necessary to achieve mode change.

(10) Furthermore, using prior art control strategies it is very difficult to simultaneously conduct both the DC-DC converter control and the DC-AC converter control to achieve mode switching. The complex procedures and simultaneous change requirement in those control strategies make it

challenging to achieve seamless mode change for grid-integration PV systems and guarantee transient switching performances.

(11) Accordingly, there is a need in the art for a unified control framework for photovoltaic systems that provides simple, seamless mode change between grid-connected mode and islanded mode as disclosed herein.

SUMMARY

(12) The following summary is provided to facilitate an understanding of some of the innovative features unique to the embodiments disclosed and is not intended to be a full description. A full appreciation of the various aspects of the embodiments can be gained by taking the entire specification, claims, drawings, and abstract as a whole.

(13) It is, therefore, one aspect of the disclosed embodiments to provide a method, system, and apparatus for energy generation.

(14) It is another aspect of the disclosed embodiments to provide a method and system for photovoltaic energy generation.

(15) It is another aspect of the disclosed embodiments to provide grid integration for photovoltaic systems.

(16) It is another aspect of the disclosed embodiments to achieve seamless mode change between grid-connected mode and islanded mode for photovoltaic systems.

(17) The aforementioned aspects and embodiments can be achieved as described herein. In an embodiment, a unified control framework for photovoltaic (PV) systems is disclosed that provides seamless mode change between grid-connected mode and islanded mode. Different from prior art control systems, in this unified control framework, the regulation of DC-bus voltage is embedded into the DC-DC converter control, and the seamless mode change is implemented in the DC-AC converter control. In this way, the DC-DC converter control and the DC-AC converter control are completely decoupled with each other. Through the DC-bus voltage control for the output voltage regulation of the DC-DC converter, a “stiff DC source” is supplied to the DC-AC converter. Then the switching between power flow control in grid-connected mode and droop control in islanded mode is implemented in the DC-AC converter control to facilitate seamless mode change. The power flow control is combined with a maximum power point tracking (MPPT) algorithm to achieve maximum power acquisition in grid-connected mode. As a result, seamless mode change in this unified control framework is achieved through three digital switches.

(18) In an embodiment a control system, comprises a PV system and a unified control framework that provides mode change between a grid-connected mode and an islanded mode. The PV system further comprises a PV collector, a DC-DC converter, and a DC-AC converter with an AC grid integration. In an embodiment the unified control framework further comprises a DC-DC converter control and a DC-AC converter control. The DC-DC converter control further comprises a voltage sensor that measures a DC-bus voltage associated with said PV system, a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal, and a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal. The DC-AC converter control further comprises at least one sensor that measures a PV output voltage, at least one sensor that measures an AC grid voltage, and at least one sensor that measures an output current of a DC-AC converter. In an embodiment the DC-AC converter control further comprises a maximum power point tracking (MPPT) unit that maximizes a power acquisition of said PV system in grid-connected mode, a real power control unit that regulates a real power and forms a control signal from a derivative of a power angle in grid-connected mode, a PV voltage regulation unit that provides a linkage between said MPPT unit and said real power control unit, and a reactive power control unit that regulates a reactive power and forms a control signal from a derivative of output voltage. In an embodiment the DC-AC converter control further comprises a P&Q calculation unit that calculates said real power and said reactive power. The DC-AC converter control further comprises a P/ω droop unit that provides a frequency regulation for an

AC grid and real power sharing in islanded mode and a Q/E droop unit that provides a voltage regulation for said AC grid and reactive power sharing in islanded mode. In an embodiment the DC-AC converter control further comprises a mode detection unit that detects an operation mode of an AC grid, three digital switch signals generated by said mode detection unit, and three digital switches configured to switch the mode of said PV system. In an embodiment the DC-AC converter control further comprises a voltage-forming unit that forms a sinusoidal control voltage according to a derivative of power angle and a derivative of output voltage. The DC-DC converter control and said DC-AC converter control are decoupled. In an embodiment the unified control framework is configured for natural DC-bus voltage protection, and a fault ride-through performance of said PV system is enhanced by DC-bus voltage protection. The mode change is implemented in said DC-AC converter control. In an embodiment, the unified control framework preserves the continuity of control signals in all control units with the configurations of digital switches.

(19) In another embodiment a system, comprises a DC-DC converter control comprising: a voltage sensor configured to measure a DC-bus voltage associated with an external system, a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal; a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal; and a DC-AC converter control comprising at least one sensor that measures an output voltage, at least one sensor that measures an AC grid voltage, and at least one sensor that measures an output current of a DC-AC converter. The DC-AC converter control further comprises a maximum power point tracking (MPPT) unit that maximizes a power acquisition in grid-connected mode, a real power control unit that regulates a real power and forms a control signal from a derivative of a power angle in grid-connected mode PV voltage regulation unit that provides a linkage between said MPPT unit and said real power control unit, and a reactive power control unit that regulates a reactive power and forms a control signal from a derivative of output voltage. The DC-AC converter control further comprises a P&Q calculation unit that calculates said real power and said reactive power, a P/ω droop unit that provides a frequency regulation for an AC grid and real power sharing in islanded mode, and a Q/E droop unit that provides a voltage regulation for said AC grid and reactive power sharing in islanded mode. In an embodiment, the DC-AC converter control further comprises three digital switch signals generated by a mode detection unit and three digital switches configured to switch the mode of said PV system.

(20) In another embodiment a control apparatus, comprises a DC-DC converter, a DC-AC converter with an AC grid integration, a voltage sensor that measures a DC-bus voltage associated with said PV system, a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal, a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal, at least one sensor that measures a PV output voltage, at least one sensor that measures an AC grid voltage, and at least one sensor that measures an output current of a DC-AC converter. In an embodiment, the mode change is implemented in said DC-AC converter control, and preserves the continuity of control signals in all control units with the configurations of digital switches.

(21) It will be appreciated that variations of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Also, it should be understood that various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, in which like reference numerals refer to identical or functionally-similar elements throughout the separate views and which are incorporated in and form a part of the specification, further illustrate the embodiments and, together with the detailed description, serve to explain the embodiments disclosed herein.

(2) FIG. **1** depicts a block diagram of a computer system which is implemented in accordance with the disclosed embodiments;

(3) FIG. **2** depicts a graphical representation of a network of data-processing devices in which aspects of the present embodiments may be implemented;

(4) FIG. **3** depicts a computer software system for directing the operation of the data-processing system depicted in FIG. **1**, in accordance with an embodiment;

(5) FIG. **4** depicts a grid-integration PV system, in accordance with a preferred embodiment;

(6) FIG. **5** depicts a boost converter, in accordance with an embodiment;

(7) FIG. **6** depicts charts of power and current as a function of voltage, in accordance with the disclosed embodiments;

(8) FIG. **7** depicts an extremum seeking (ES) method, in accordance with the disclosed embodiments;

(9) FIG. **8** depicts an electric circuit of a DC-AC converter, in accordance with the disclosed embodiments;

(10) FIG. **9** depicts an exemplary unified control system, in accordance with the disclosed embodiments;

(11) FIG. **10**A depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(12) FIG. **10**B depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(13) FIG. **10**C depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(14) FIG. **10**D depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(15) FIG. **10**E depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(16) FIG. **10**F depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(17) FIG. **11**A depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments;

(18) FIG. **11**B depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments;

(19) FIG. **11**C depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments;

(20) FIG. **11**D depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments;

(21) FIG. **12**A depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(22) FIG. **12**B depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(23) FIG. **12**C depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(24) FIG. **12**D depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(25) FIG. **12**E depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(26) FIG. **12**F depicts a chart of exemplary results, in accordance with the disclosed embodiments;

(27) FIG. **13**A depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments;

(28) FIG. **13**B depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments;

(29) FIG. **13**C depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments; and

(30) FIG. **13**D depicts a chart of transient performances from grid-connected mode to islanded mode, in accordance with the disclosed embodiments.

DETAILED DESCRIPTION

(31) The particular values and configurations discussed in the following non-limiting examples can be varied and are cited merely to illustrate one or more embodiments and are not intended to limit the scope thereof.

(32) Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which illustrative embodiments are shown. The embodiments disclosed herein can be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the embodiments to those skilled in the art. Like numbers refer to like elements throughout.

(33) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprise” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(34) Throughout the specification and claims, terms may have nuanced meanings suggested or implied in context beyond an explicitly stated meaning. Likewise, the phrase “in one embodiment” as used herein does not necessarily refer to the same embodiment and the phrase “in another embodiment” as used herein does not necessarily refer to a different embodiment. It is intended, for example, that claimed subject matter include combinations of example embodiments in whole or in part.

(35) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(36) It is contemplated that any embodiment discussed in this specification can be implemented with respect to any method, kit, reagent, or composition of the invention, and vice versa. Furthermore, compositions of the invention can be used to achieve methods of the invention.

(37) It will be understood that particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention can be employed in various embodiments without departing from the scope of the invention. Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation, numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.

(38) The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.” The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.” Throughout this application, the term “about” is used to indicate that a value includes the inherent variation of error for the device, the method being employed to determine the value, or the variation that exists among the study subjects.

(39) As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps.

(40) The term “or combinations thereof” as used herein refers to all permutations and combinations of the listed items preceding the term. For example, “A, B, C, or combinations thereof” is intended to include at least one of: A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, CBA, BCA, ACB, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB, AAA, AB, BBC, AAABCCCC, CBBAAA, CABABB, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.
(41) All of the compositions and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit, and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the invention as defined by the appended claims.
(42) FIGS. **1-3** are provided as exemplary diagrams of data-processing environments in which embodiments of the present invention may be implemented. It should be appreciated that FIGS. **1-3** are only exemplary and are not intended to assert or imply any limitation with regard to the environments in which aspects or embodiments of the disclosed embodiments may be implemented. Many modifications to the depicted environments may be made without departing from the spirit and scope of the disclosed embodiments.
(43) A block diagram of a computer system **100** that can serve as a control device that executes programming for implementing parts of the methods and systems disclosed herein is shown in FIG. **1**. A computing device in the form of a computer **110** configured to interface with controllers, peripheral devices, and other elements disclosed herein may include one or more processing units **102**, memory **104**, removable storage **112**, and non-removable storage **114**. Memory **104** may include volatile memory **106** and non-volatile memory **108**. Computer **110** may include or have access to a computing environment that includes a variety of transitory and non-transitory computer-readable media such as volatile memory **106** and non-volatile memory **108**, removable storage **112** and non-removable storage **114**. Computer storage includes, for example, random access memory (RAM), read only memory (ROM), erasable programmable read-only memory (EPROM) and electrically erasable programmable read-only memory (EEPROM), flash memory or other memory technologies, compact disc read-only memory (CD ROM), Digital Versatile Disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices, or any other medium capable of storing computer-readable instructions as well as data including image data.
(44) Computer **110** may include, or have access to, a computing environment that includes input **116**, output **118**, and a communication connection **120**. The computer may operate in a networked environment using a communication connection **120** to connect to one or more remote computers, remote sensors and/or controllers, detection devices, hand-held devices, multi-function devices (MFDs), speakers, mobile devices, tablet devices, mobile phones, Smartphone, or other such devices. The remote computer may also include a personal computer (PC), server, router, network PC, RFID enabled device, a peer device or other common network node, or the like. The communication connection may include a Local Area Network (LAN), a Wide Area Network (WAN), Bluetooth connection, or other networks. This functionality is described more fully in the description associated with FIG. **2** below.
(45) Output **118** is most commonly provided as a computer monitor, but may include any output device. Output **118** and/or input **116** may include a data collection apparatus associated with control device **100**. In addition, input **116**, which commonly includes sensor signals from control targets

**135**, e.g., voltages, currents, etc. from electrical systems, but may also include a computer keyboard and/or pointing device such as a computer mouse, computer track pad, or the like, allows a user to select and instruct control device **100**. Output **118** commonly includes the control signals acted on actuators **140**, e.g., Pulse Width Modulation (PWM) signals for powering electronic devices in electrical systems, but also may include a display for displaying data and information for a user. A user interface can be provided using output **118** and input **116** for interactively displaying a graphical user interface (GUI) **130**.

(46) Note that the term “GUI” generally refers to a type of environment that represents programs, files, options, and so forth by means of graphically displayed icons, menus, and dialog boxes on a computer monitor screen. A user can interact with the GUI to select and activate such options by directly touching the screen and/or pointing and clicking with a user input device **116** such as, for example, a pointing device such as a mouse, and/or with a keyboard. A particular item can function in the same manner to the user in all applications because the GUI provides standard software routines (e.g., module **125**) to handle these elements and report the user's actions. The GUI can further be used to display the electronic service image frames as discussed below.

(47) Computer-readable instructions, for example, program module or node **125**, which can be representative of other modules or nodes described herein, are stored on a computer-readable medium and are executable by the processing unit **102** of computer control device **110**. Program module or node **125** may include a computer application. A hard drive, CD-ROM, RAM, Flash Memory, and a USB drive are just some examples of articles including a computer-readable medium.

(48) FIG. **2** depicts a graphical representation of a network of data-processing systems **200** in which aspects of the present invention may be implemented. Network data-processing system **200** can be a network of computers or other such devices, such as mobile phones, smart phones, sensors, controllers, speakers, tactile devices, and the like, in which embodiments of the present invention may be implemented. Note that the system **200** can be implemented in the context of a software module such as program module **125**. The system **200** includes a network **202** in communication with one or more clients **210**, **212**, and **214**. Network **202** may also be in communication with one or more devices **204**, servers **206**, and storage **208**. Network **202** is a medium that can be used to provide communications links between various devices and computers connected together within a networked data processing system such as control device **100**. Network **202** may include connections such as wired communication links, wireless communication links of various types, and fiber optic cables. Network **202** can communicate with one or more servers **206**, one or more external devices such as device **204**, and a memory storage unit such as, for example, memory or database **208**. It should be understood that device **204** may be embodied as a PV system, or subcomponent of a PV system as disclosed herein.

(49) In the depicted example, device **204**, server **206**, and clients **210**, **212**, and **214** connect to network **202** along with storage unit **208**. Clients **210**, **212**, and **214** may be, for example, personal computers or network computers, handheld devices, mobile devices, tablet devices, smart phones, personal digital assistants, printing devices, recording devices, speakers, MFDs, etc. Computer system **100** depicted in FIG. **1** can be, for example, device **204** may be embodied as a PV system, or a client such as client **210** and/or **212** and/or **214**.

(50) Computer system **100** can also be implemented as a server such as server **206**, depending upon design considerations. In the depicted example, server **206** provides data such as boot files, operating system images, applications, and application updates to clients **210**, **212**, and/or **214**. Clients **210**, **212**, and **214** and device **204** are clients to server **206** in this example. Network data-processing system **200** may include additional servers, clients, and other devices not shown. Specifically, clients may connect to any member of a network of servers, which provide equivalent content.

(51) In the depicted example, network data-processing system **200** is the Internet, with network **202**

representing a worldwide collection of networks and gateways that use the Transmission Control Protocol/Internet Protocol (TCP/IP) suite of protocols to communicate with one another. At the heart of the Internet is a backbone of high-speed data communication lines between major nodes or host computers consisting of thousands of commercial, government, educational, and other computer systems that route data and messages. Of course, network data-processing system **200** may also be implemented as a number of different types of networks such as, for example, an intranet, a local area network (LAN), or a wide area network (WAN). FIGS. **1** and **2** are intended as examples and not as architectural limitations for different embodiments of the present invention.

(52) FIG. **3** illustrates a software system **300**, which may be employed for directing the operation of the data-processing systems such as control device **100** depicted in FIG. **1**. Software application **305**, may be stored in memory **104**, on removable storage **112**, or on non-removable storage **114** shown in FIG. **1**, and generally includes and/or is associated with a kernel or operating system **310** and a shell or interface **315**. One or more application programs, such as module(s) or node(s) **125**, may be "loaded" (i.e., transferred from removable storage **114** into the memory **104**) for execution by the data-processing system **100**. The data-processing system **100** can receive user commands and data through user interface **315**, which can include input **116** and output **118**, accessible by a user **320**. These inputs may then be acted upon by the control device or computer system **100** in accordance with instructions from operating system **310** and/or software application **305** and any software module(s) **125** thereof.

(53) Generally, program modules (e.g., module **125**) can include, but are not limited to, routines, subroutines, software applications, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types and instructions. Moreover, those skilled in the art will appreciate that elements of the disclosed methods and systems may be practiced with other computer system configurations such as, for example, hand-held devices, mobile phones, smart phones, tablet devices multi-processor systems, microcontrollers, printers, copiers, fax machines, multi-function devices, data networks, microprocessor-based or programmable consumer electronics, networked personal computers, minicomputers, mainframe computers, servers, medical equipment, medical devices, and the like.

(54) Note that the term "module" or "node" as utilized herein may refer to a collection of routines and data structures that perform a particular task or implements a particular abstract data type. Modules may be composed of two parts: an interface, which lists the constants, data types, variables, and routines that can be accessed by other modules or routines; and an implementation, which is typically private (accessible only to that module), and which includes source code that actually implements the routines in the module. The term module may also simply refer to an application such as a computer program designed to assist in the performance of a specific task such as word processing, accounting, inventory management, etc., or a hardware component designed to equivalently assist in the performance of a task.

(55) The interface **315** (e.g., a graphical user interface **130**) can serve to display results, whereupon a user **320** may supply additional inputs or terminate a particular session. In some embodiments, operating system **310** and GUI **130** can be implemented in the context of a "windows" system. It can be appreciated, of course, that other types of systems are possible. For example, rather than a traditional "windows" system, other operation systems such as, for example, a real-time operating system (RTOS) more commonly employed in wireless systems may also be employed with respect to operating system **310** and interface **315**. The software application **305** can include, for example, module(s) **125**, which can include instructions for carrying out steps or logical operations such as those shown and described herein.

(56) The following description is presented with respect to embodiments of the present invention, which can be embodied in the context of, or require the use of, a data-processing system such as control device **100**, in conjunction with program module **125**, and data-processing system **200** and network **202** depicted in FIGS. **1-3**. The present invention, however, is not limited to any particular

application or any particular environment. Instead, those skilled in the art will find that the system and method of the present invention may be advantageously applied to a variety of system and application software including database management systems, word processors, and the like. Moreover, the present invention may be embodied on a variety of different platforms including Windows, Macintosh, UNIX, LINUX, Android, advanced RISC machine (ARM), digital signal processor (DSP), Arduino and the like. Therefore, the descriptions of the exemplary embodiments, which follow, are for purposes of illustration and not considered a limitation.
(57) The embodiments disclosed herein present a unified control framework for photovoltaic (PV) systems to achieve seamless mode change between grid-connected mode and islanded mode. Different from traditional designs, in this unified control framework, the regulation of DC-bus voltage is embedded into the DC-DC converter control, and the seamless mode change is implemented in the DC-AC converter control. In this way, the DC-DC converter control and the DC-AC converter control can be completely decoupled from each other. Through the DC-bus voltage control for the output voltage regulation of the DC-DC converter, a “stiff DC source” is supplied to the DC-AC converter. Then the switching between power flow control in grid-connected mode and droop control in islanded mode is implemented in the DC-AC converter control to facilitate seamless mode change. The power flow control is combined with a maximum power point tracking (MPPT) algorithm to achieve maximum power acquisition in grid-connected mode. As a result, seamless mode change in this unified control framework can be provided through three digital switches.
(58) FIG. **4**, illustrates a grid-integration PV system **400** with dual-stage converters in accordance with the disclosed embodiments. The system **400** includes a PV source **405** (e.g., photovoltaic solar panels), a DC-DC converter **410** and a DC-AC converter **415**. The PV source **405** and the DC-DC converter **410** are connected in series with a capacitor C.sub.pv **420** to stabilize the PV output voltage. The DC-DC converter **410** and the DC-AC converter **415** are connected in series with another capacitor C.sub.dc or a DC-bus **425**, to smooth the DC-bus voltage. The AC grid **430** can be the utility grid or the microgrid. The system **400** and associated methods disclosed herein provide a unified control framework that allows seamless mode change.
(59) The unified control system **400**, illustrated in FIG. **4**, includes two decoupled modules: the DC-DC converter control **435** and the DC-AC converter control **440**. The DC-DC converter control **435** includes a DC-bus voltage control unit **445** and a pulse width modulation (PWM) generation unit **450**. Through the DC-bus voltage control **445**, the variable-voltage DC power directly generated from the PV source **405** is converted to the constant-voltage DC-bus **425**. The PWM generation unit **450** transfers the duty-cycle output of the DC-bus voltage control unit **445** into the final PWM control signal for the DC-DC converter **410**.
(60) The DC-AC converter control **440** includes a maximum power point tracking (MPPT) unit **455**, a P&Q calculation unit **460**, a PV voltage control unit **465**, a mode detection unit **470**, two power flow control units (including real power control **475** and reactive power control **476**), two droop control units (including P/ω droop control unit **480** and Q/E droop control unit **481**), a sinusoidal voltage-forming unit **485**, and a PWMs generation unit **490**.
(61) The P&Q calculation unit **460** is adopted to calculate both real power output P.sub.g and reactive power output Q.sub.g of the DC-AC converter **415**. The real power output P.sub.g will be maximized through the MPPT unit **455**, and the power flow control units (i.e., real power control **475** and reactive power control **476**) convert DC power from the DC-bus **425** into AC grid **430** in grid-connected mode. The PV voltage control unit **465** is configured to bridge the MPPT unit **455** and the real power control **475**.
(62) The mode detection unit **470** is used for detecting the grid operation conditions. Note that, in certain embodiments, the mode detection unit **470** could be embodied as external trigger signals, e.g., from a microgrid centralized controller (MGCC). The droop control units **480** and **481** provide frequency regulation, voltage regulation, and power sharing in islanded mode. Three digital

switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497** associated with mode detection unit **470**, are used to change the operation modes of the PV system **400**. The sinusoidal voltage-forming unit **485** combines the amplitude and frequency of the control voltage, and the PWMs generation unit **490** generates the final PWMs control signals for the DC-AC converter **415**.
(63) In the unified control system **400**, the regulation of the DC-bus voltage is implemented in the DC-DC converter control **435**, and the mode change is integrated in the DC-AC converter control **440**, as shown in FIG. **4**. In an embodiment, the DC-DC converter control **435** and the DC-AC converter control **440** are decoupled. Thus, the DC-bus voltage regulation will be always guaranteed, and is not affected by the behaviors in the DC-AC converter control **440** (e.g., mode change). The unified control system **400** naturally provides the DC-bus voltage protection, even in the event of fault conditions in the DC/AC converter **415** or AC grid side **430**. Through the DC-bus voltage control **445** in the DC-DC converter control **435**, a “stiff DC source” can be reliably supplied to the DC-AC converter **415**. Therefore, seamless mode change can be achieved in the DC-AC converter control **440**. Also, the fault ride-through performance of the whole system **400** is enhanced by DC-bus voltage protection.
(64) The three switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497** offer further advantages associated with the system **400**. The seamless mode change in the DC-AC converter control **440** for the system **400** is achieved via operating switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497**.
(65) Two operation modes are illustrated in Table 1.
(66) TABLE-US-00001 TABLE 1 OPERATION MODES OF GRID-INTEGRATION PV SYSTEMS Switch S.sub.P Switch S.sub.W Switch S.sub.Q Mode 1 1 1 Grid-connected mode 2 2 2 Islanded mode
(67) When the switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497** switch to position 1, as shown in Table 1, the system **400** operates in grid-connected mode. The DC-AC converter control **440** includes four basic steps: MPPT **455**, PV voltage control **465**, power flow control of both real power **475** and reactive power **476**, and sinusoidal control voltage generation **485**. Here, MPPT **455**, which can be embodied as a P&O algorithm, is focused on the output power P.sub.g to achieve a maximum system output.
(68) The perturbation signal of MPPT **455** cannot be added on the real power reference P*.sub.g directly, because the positive perturbation of P*.sub.g at the maximum point might cause system instability. The PV voltage control unit **465** can be used to mitigate this effect. The reactive power reference Q*.sub.g is usually set to zero with Q.sub.set=0 to keep the unity power factor in grid-connected mode. The sinusoidal control voltage v.sub.r is generated according to the derivatives of both power angle and output voltage passing through the integration units and combining with the global settings.
(69) When the switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497** switch to position 2 shown in Table 1, the system **400** operates at islanded mode with droop control. The P/ω droop unit **480** provides frequency regulation in the real power channel. The Q/E droop unit **481** generates the reactive power reference Q*.sub.g, which combines with the reactive power control **476** to provide voltage regulation in the reactive power channel. Accordingly, the power sharing of both real power and reactive power can be guaranteed.
(70) In this unified control framework, only three switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497**, are necessary in the DC-AC converter control **440** for mode change between grid-connected mode and islanded mode. It is worth noting that the major function of switch S.sub.P **495** is to preserve the continuity of the real power control unit **475** in islanded mode. If S.sub.P **495** switches to position 2, it can prevent the signal {dot over (δ)} from reaching infinity even in the presence of the mismatched real power reference P*.sub.g and real power input P.sub.g in islanded mode. S.sub.P **495** is also used to enable/disable both MPPT unit **455** and PV voltage control unit **465** in different modes.
(71) The switches S.sub.W **496** and S.sub.Q **497** provide mode changes. In this way, the unified

control framework achieves seamless mode change for PV systems. It is noteworthy that this unified control framework has a simple design and does not require any reconfiguration of the control systems. As a result, a simpler procedure is enabled by the operation of switches S.sub.P **495**, S.sub.W **496**, and S.sub.Q **497**.

(72) Though all control units in the proposed control system **400** in FIG. **4** can be implemented using any number of algorithms, the framework itself, especially, the DC-bus voltage control **445** in the DC-DC converter control **435**, and mode change in the DC-AC converter control **440**, are of particular note. The detailed control algorithms for the unified control framework to facilitate seamless mode change are incorporated as embodiments disclosed herein.

(73) With respect to DC-bus voltage control, in certain grid-integration PV systems, a DC-DC boost converter can be used. FIG. **5** illustrates a boost converter **500** in accordance with an embodiment. It should be understood that in certain embodiments other types of DC-DC converters can be applied according to design consideration. The model of the boost converter without parasitics is represented according to equations (1) and (2) as follows:

(74) $\dot{V}_{dc} = (1-u)\frac{i_L}{C_{dc}} - \frac{V_{dc}}{C_{dc}R_v}$ (1) $\dot{i}_L = -(1-u)\frac{V_{dc}}{L} + \frac{V_{pv}}{L} - \frac{R_L}{L}i_L$ (2) where V.sub.pv is the input voltage, which is connected to the output of the PV source **405**; V.sub.dc is the output voltage, which is connected to the input of the DC-AC converter **415**; i.sub.L is the current of the inductor **505**; L is the inductor **505** inductance and R.sub.L is the inductor **505** resistance; C.sub.pv **420** is the capacitance of the input capacitor and C.sub.dc **425** is the output capacitance; R.sub.v is the virtual output resistance **510** to represent the load to feed the DC-AC converter **415**, and R.sub.v changes according to different power consumptions of the DC-AC converter **415**; u is the input within the range of 0-100%.

(75) For the DC-bus voltage control of the DC-DC boost converter **500**, many robust control designs based on the dynamics of equations (1) and (2) can be adopted. For example, in an embodiment, a PI controller, plus feed-forward terms, as shown in equation (3) can be adopted:

(76) $u = 1 - \frac{V_{pv}}{V^*_{dc}} + K_{pdc}(V^*_{dc} - V_{dc}) + K_{idc}\int(V^*_{dc} - V_{dc})dt$ (3) where V*.sub.dc is a DC-bus voltage reference, and K.sub.pdc>0 and K.sub.idc>0 are PI control gains.

(77) The PV output power and output current vary greatly with different loads and different sunlight conditions. According to a widely used five-parameter PV model, the i.sub.pv-V.sub.pv and P.sub.pv-V.sub.pv fitting curves for a commercial solar panel, 50 W RENOGY RNG-50P at a standard test temperature T=25° C. are shown in FIG. **6** at different sunlight irradiation levels. The current output and the power output under different sunlight irradiation levels and different output voltages (or load conditions) are illustrated in chart **605** and chart **610**, respectively.

(78) It is noteworthy that the PV source **405** will have many operating points, according to different loads under different environmental conditions. Thus, an optimal tool is required to get maximum power output of PV systems. In certain embodiments, the gradient-based extremum seeking (ES) algorithm can be used to maximize the real power output P.sub.g of grid-integration PV system **400** for the disclosed unified control framework.

(79) An evolution strategy (ES) method **700** is illustrated in FIG. **7**. In the embodiment, the perturbation signal **705** α sin(Ω.sub.pt) is injected, for the observation of ĝ **725** to achieve the maximum power output **710** P.sub.g, where

(80) $\frac{s}{s+\ \ H}$

is a high-pass filter **715** to filter the DC part of P.sub.g,

(81) $\frac{L}{s+\ \ L}$

is a low-pass filter **720** to filter the double frequency of the perturbation signal, and K.sub.es>0 is the integral or incremental gain for ĝ.

(82) The PV voltage control unit **465** (shown in FIG. **4**) can be designed with a PI controller, which can be characterized by equation (4):

(83) $P_g^* = -K_{ppv}(V_{pv}^* - V_{pv}) - K_{ipv}\int(V_{pv}^* - V_{pv})dt$ (4) where V*.sub.pv is from the MPPT unit **455**, and K.sub.ppv>0 and K.sub.ipv>0 are PI gains. Note that the PI controller for the PV voltage control **465** has negative gains. These negative gains are caused by the negative slope of P.sub.pv-V.sub.pv curves of the solar panel from the open circuit point to the maximum power output point (as illustrated in chart **610** of FIG. **6**).

(84) Power flow control is also necessary. FIG. **8** illustrates an embodiment of the electric circuit **800** of a DC-AC converter **415**, where the power delivered from converter side E∠δ **805** to AC grid U.sub.g∠0° **810** is calculated according to equations (5) and (6):

(85) $P_g = (\frac{E_g U_g}{Z_g}\cos \quad - \frac{U_g^2}{Z_g})\cos \quad + \frac{E_g U_g}{Z_g}\sin \quad \sin$ (5)

$Q_g = (\frac{E_g U_g}{Z_g}\cos \quad - \frac{U_g^2}{Z_g})\sin \quad - \frac{E_g U_g}{Z_g}\sin \quad \cos$ (6)

(86) In equations (5) and (6), P.sub.g is real power, Q.sub.g is reactive power, and δ is power angle; output impedance is represented as shown in equation (7)

(87) $Z_g \angle \theta = R_g + X_g j$ (7)

(88) In equation (7), X.sub.g is determined primarily by inductor **815** L.sub.g, because C.sub.g usually can be neglected; CB is an internal circuit breaker **820** for the DC-AC converter **415**.

(89) By taking derivatives of equations (5) and (6), the dynamics of power delivery can be obtained as shown in equations (8) and (9):

(90) $\dot{P}_g = \frac{E_g U_g}{Z_g}\dot{\ } + \triangle_p$ (8) $\dot{Q}_g = \frac{U_g}{Z_g}\dot{E}_g + \triangle_q$ (9)

where Δ.sub.p and Δ.sub.q are given in equations (10) and (11) as follows:

(91) $\triangle_p = \frac{E_g U_g}{Z_g}\dot{\ }(\cos \quad \sin \quad - 1) - \frac{E_g U_g}{Z_g}\dot{\ }\sin \quad \cos \quad + \frac{U_g \dot{E}_g}{Z_g}\cos \quad \cos \quad + \frac{U_g \dot{E}_g}{Z_g}\sin \quad \sin$ (10)

$\triangle_q = \frac{U_g \dot{E}_g}{Z_g}(\cos \quad \sin \quad - 1) - \frac{U_g \dot{E}_g}{Z_g}\sin \quad \sin \quad - \frac{E_g U_g}{Z_g}\dot{\ }\cos \quad \cos \quad - \frac{U_g \dot{E}_g}{Z_g}\sin \quad \cos$ (11)

(92) Δ.sub.p and Δ.sub.q indicate lumped unknown terms, which include uncertainties, nonlinearities, and coupling effects of power angle δ and output impedance Z.sub.g∠θ.

(93) According to the robust control methods disclosed herein, an uncertainty and disturbance estimator (UDE)-based control, and the power flow control with both real power control and reactive power control can be given according to equations (12) and (13) as:

(94) $0 \dot{\ } = \frac{Z_g}{E_g U_g}[L^{-1}\{\frac{1}{1 - G_{pf}(s)}\} * (\dot{P}_g^* + k_p e_p) - L^{-1}\{\frac{sG_{pf}(s)}{1 - G_{pf}(s)}\} * P_g]$ (12)

$\dot{E}_g = \frac{Z_g}{U_g}[L^{-1}\{\frac{1}{1 - G_{qf}(s)}\} * (\dot{Q}_g^* + k_q e_q) - L^{-1}\{\frac{sG_{qf}(s)}{1 - G_{qf}(s)}\} * Q_g]$ (13)

(95) In equations (12) and (13), P*.sub.g is a real power reference, and Q*.sub.g is a reactive power reference. In grid-connected mode, P*.sub.g is from the PV voltage control unit **465**, and Q*.sub.g is usually set to zero with Q.sub.set=0 to keep unity power factor. In islanded mode, P*.sub.g is from real power P.sub.g to prevent the signal {dot over (δ)} from reaching infinity, and Q*.sub.g is from Q/E droop unit **481**. e.sub.p=P*.sub.g−P.sub.g and e.sub.q=Q*.sub.g−Q.sub.g are tracking errors. k.sub.p>0 and k.sub.q>0 are error feedback gains for error dynamics equations, ė.sub.p=−k.sub.pe.sub.p and ė.sub.q=−k.sub.qe.sub.q in the UDE-based control. G.sub.pf(s) and G.sub.qf(s) are UDE filters to estimate the unknown terms Δ.sub.p in equation (10) and Δ.sub.q in equation (11).

(96) The droop control is primarily used for islanded mode operation of renewable generation units, and droop control can be implemented as illustrated in FIG. **4**. The frequency regulation with P/ω droop **480** can be designed according to equation (14) as:

(97) $= {}^* - mP_g$ (14)

(98) In equation (14), m is the real power droop coefficient, and ω*=2πf*, and f* are the rated frequency. In the steady-state, all parallel units in an islanded microgrid should hold the same frequency ω. When that is the case, mP.sub.g in the PV system **400** is equal to other parallel units, which guarantees the accurate sharing of real power from the PV system **400** with other parallel units.
(99) For Q/E droop **481**, the reactive power reference Q*.sub.g is generated according to equation (15):
(100) $Q_g^* = \frac{E^* - U_g}{n}$ (15)
(101) In equation (15), E* is the rated voltage, U.sub.g is the root-mean-square (RMS) value of the instantaneous load voltage u.sub.g, and n is the reactive power droop coefficient. Combining the reactive power control in equation (13), this provides Q/E droop with voltage regulation. In the steady-state, reactive power output Q.sub.g will track the reference Q*.sub.g through the reactive power control (13) according to equation (16):
(102) $Q_g = Q_g^* = \frac{E^* - U_g}{n}$ (16)
(103) Then nQ.sub.g in the PV system **400** can be equal to those in other parallel units, because both E* and U.sub.g are the same for all parallel-operated units. This guarantees accurate reactive power sharing of the PV system **400** with other parallel units.
(104) Mode detection **470** is a critically important aspect of the embodiments disclosed herein. When the microgrid is disconnected from the utility grid, a mode detection unit needs to trigger the islanded mode operation of the PV system **400** with droop control.
(105) There are two kinds of islanded detection methods: passive techniques and active techniques. Passive techniques are generally based on monitoring and processing local signals (e.g., grid voltage, output current, or grid frequency).
(106) In certain embodiments, passive methods include detections of over-/under-voltage, detections of over-/under-frequency, detections of voltage/current harmonics, and state estimations. The active techniques usually introduce some disturbances to the system output intentionally and analyze the behaviors of the system responses to these disturbances to determine the system's operational conditions. In other embodiments, major active methods include harmonics injection/detection of impedance, variation of real power and reactive power, etc. As a general rule, active islanded detection techniques are more accurate than passive techniques, but also more complex. Active methods also introduce harmonics into the grid system. In certain embodiments mode change also can be triggered by external signals with other requirements, e.g., from a microgrid centralized controller (MGCC).
(107) An exemplary unified control system **900** for seamless mode change includes a PV system with both grid-connected mode and islanded mode, as shown in FIG. **9**. The system **900** was used to test the embodiments disclosed herein, where the PV system **900** with some local RC loads **920** are connected with the utility grid **925** through a circuit breaker CB1 **910**. The solar sources **405** are designed with the 10×10 series-parallel arrays of RENOGY RNG-50P. RC loads **920** can comprise 14.4Ω resistance and 80 μF capacitance. The utility grid **925** can comprise a three-phase AC grid with rated phase-voltage amplitude of 120 V.sub.rms and rated frequency of 60 Hz. When both CB1 **910** and circuit breaker CB2 **915** are ON, the PV system **900** operates at grid-connected mode. When CB2 **915** is turned OFF, the PV system **900** operates at islanded mode. An additional grid voltage measurement {dot over (u)}.sub.g **930** is conducted by the DC-AC converter **415** for initial synchronization, when the PV system **900** switches from islanded mode to grid-connected mode. System parameters associated with this configuration are provided in Table 2, and control parameters are given in Table 3. The solar sources **405** are assumed to work at a standard test environment.
(108) TABLE-US-00002 TABLE 2 System Parameters Parameters Values Parameters Values

$C_{pv}$ 1000 μF $L_g$ 2.2 mH L 100 μH $R_g$ 0.5 Ω $R_L$ 0.1 Ω $C_g$ 10 μF $C_{dc}$ 2000 μF Nominal grid voltage $E^*/\sqrt{3}$ 120 $V_{rms}$ $V_{dc}^*$ 400 V Nominal grid frequency f* 60 Hz
(109) TABLE-US-00003 TABLE 3 Control Parameters Parameters Values Parameters Values $K_{pdc}$ 0.08 $K_{ipv}$ 100 $k_{idc}$ 0.2 $k_p$ 10 $\Omega_p$ 10π rad/s $k_q$ 2 a 2 $G_{pf}(s)$ $\frac{40}{s+40}$ $k_{es}$ 0.5 $G_{qf}(s)$ $\frac{8}{s+8}$ $\Omega_p$ 2π rad/s m 4π * $10^{-5}$ $\Omega_p$ 5π rad/s n $0.002\sqrt{3}$ $K_{ppv}$ 20 — —
(110) In one case, the system **900** can move from grid-connected mode to islanded mode. The system **900** starts in grid-connected mode with both CB1 **910** and CB2 **915** ON. Initially, the DC-DC converter **410** regulates the DC-bus voltage. Next, the DC-AC converter **415** synchronizes with the utility grid **925** through the internal circuit breaker CB, shown in FIG. **8**, and starts to deliver a small amount of power to the utility grid **925**. The MPPT **455** can be enabled at t=1 s. At t=20 s, CB2 **915** can be turned OFF to mimic sudden loss of the utility grid **925**, then the PV system **900** is forced to operate in islanded mode. Here, a simple mode detection based on the measurement of load voltage $U_g$ can be used. If $U_g$ is out of the range $115\sqrt{3}$ $V_{rms} \leq U_g \leq 125\sqrt{3}$ $V_{rms}$, then islanded mode can be triggered, and CB1 **910** is also turned OFF.
(111) The results of this test case are shown in FIGS. **10**A-**10**F. After the system starts, the DC-bus voltage is well regulated to 400 V, as shown in chart **1030** in FIG. **10**F. When the MPPT **455** is enabled, the real power output increases gradually to the maximum value at about t=17 s, as shown in chart **1005** in FIG. **10**A. The frequency is around the grid frequency of 60 Hz, with some oscillations as shown in chart **1010** in FIG. **10**B. These oscillations are caused by the ES-based MPPT **455** due to the oscillations of real power. The oscillations of the frequency decrease gradually, as real power gradually approaches the maximum value. The reactive power output is controlled to 0 Var to maintain the unity power factor, and the load voltage is the same as grid voltage of 120 $V_{rms}$, as shown in chart **1015** FIG. **10**C and chart **1020** FIG. **10**D, respectively. PV voltage decreases gradually, because more real power is generated from solar panels with the MPPT **455**, as shown in chart **1025** in FIG. **10**E. At t=20 s, the utility grid **925** is lost. The load voltage surges quickly in FIG. **10**D.
(112) After the islanded condition is detected, the PV system **900** switches to islanded mode. With several cycles of transient-states, the whole system **900** converges to a steady-state quickly. The frequency is well regulated with P/ω droop control **480**, as illustrated in FIG. **10**B, and the real power drops to about 3.1 kW as shown in FIG. **10**A. The small drop in frequency is caused by positive real power with resistance load. Also, the load voltage is well regulated with Q/E droop control **481** in FIG. **10**D, and the reactive power is around −1.4 kVar which supports the capacitance load as shown in FIG. **10**C. The small increase in load voltage is caused by negative reactive power. The PV voltage increases to a constant level with less constant real power output of the solar panels as shown in FIG. **10**E, and the DC-bus voltage is still well regulated as shown in chart **1030** in FIG. **10**F.
(113) The transient performances from grid-connected mode to islanded mode is shown in FIGS. **11**A-**11**D. The AC currents have one-cycle distortions because of the sudden loss of the utility grid, but they converge to smooth curves quickly, as shown in chart **1115** FIG. **11**C. The current distortions cause some voltage distortions in chart **1120** in FIG. **11**D. The distortions of both currents and voltages also cause the overshoots in both real power and reactive power, but they converge to steady-states soon, as shown in chart **1105** in FIG. **11**A and chart **1110** in FIG. **11**B, respectively. Therefore, the proposed unified control framework can provide seamless mode change from grid-connected mode to islanded mode.
(114) In another case, the system **900** can move from islanded mode to grid-connected mode. The system starts in islanded mode with CB1 **910** in the OFF position and CB2 **915** in the OFF position. Initially, the DC-DC converter **410** regulates the DC-bus voltage, then the DC-AC

converter **415** starts to deliver power to the RC loads **920** and regulate both frequency and load voltage as well. At t=10 s, CB2 **915** can be turned ON, and the PV system **900** starts to synchronize with the utility grid **925** through the measurement of grid voltage {dot over (u)}.sub.g. At t=10.05 s, CB1 **910** is turned ON, and the PV system **900** switches to grid-connected mode with both MPPT **455** and power flow control **475** and **476** at t=10.1 s.

(115) The results of this test case are illustrated in FIGS. **12**A-**12**F. From chart **1210** in FIG. **12**B and chart **1220** in FIG. **12**D, it is clear that both frequency and load voltage are well regulated (after some small overshoots) in islanded mode. Both real power in chart **1205** FIG. **12**A and reactive power in chart **1215** FIG. **12**C converge to steady-states quickly with the droop control **480** and **481**. The DC-bus voltage is well regulated as shown in chart **1230** in FIG. **12**F, and the PV voltage keeps constant with constant real power output as illustrated in chart **1225** in FIG. **12**E. The whole system demonstrates similar steady-state performance to the first case in islanded mode. When CB2 **915** is turned ON at t=10 s, with the procedures of the initial synchronization and CB1 **910** turned ON, the PV system **900** goes to grid-connected mode with both MPPT **455** and power flow control **475** and **476**. The real power drops when CB1 **910** is ON, because the utility grid **925** shares some real power to support the loads. Then real power gradually increases to the maximum value with the MPPT **455**, as shown in FIG. **12**A. The frequency increases to the grid frequency of 60 Hz as illustrated in FIG. **12**B, and it has some oscillations with the same reason of the first case. The reactive power in FIG. **12**C converges to 0 Var, because Q.sub.set=0 Var. FIG. **12**D shows that the load voltage drops to the grid voltage of 120 V.sub.rms, because the load is directly connected to the utility grid **925**. The PV voltage decreases accordingly because of the MPPT **455** as shown in FIG. **12**E, while the DC-bus voltage is still well controlled as shown in FIG. **12**F through the DC-bus voltage control **445**.

(116) The transient performances from islanded mode to grid-connected mode are shown in FIGS. **13**A-**13**D. After t=10 s, the DC-AC converter **415** of the PV system **900** starts to synchronize with utility grid **925** through {dot over (u)}.sub.g. The real power in chart **1305** FIG. **13**A has small drops and reactive power in chart **1310** in FIG. **13**B has small increases, because load voltage reduces to grid voltage through grid synchronization. After the initial synchronization and CB1 **910** are turned ON at t=10.05 s, there are some transient-states in both real power and reactive power, but they converge to steady-states quickly. After t=10.1 s, real power starts to increase with the MPPT **455**, and reactive power starts to converge to 0 Var. There are almost no current distortions and voltage distortions as illustrated by chart **1315** in FIG. **13**C and chart **1320** in FIG. **13**D. Therefore, the unified control framework can provide seamless mode change from islanded mode to grid-connected mode.

(117) In certain embodiments the methods and systems disclosed herein can be characterized as a control system, comprising a class of grid-integration PV systems and a unified control framework that allows for seamless mode change between grid-connected mode and islanded mode. The grid-integration PV systems can comprise a PV source, a DC-DC converter, and a DC-AC converter with AC grid integration. The unified control framework is configured for seamless mode change between grid-connected mode and islanded mode. The unified control framework can further comprise a DC-DC converter control and a DC-AC converter control.

(118) The control system associated with the DC-DC converter control is configured for controlling the DC-DC converter in the grid-integration PV system. The control system can include a voltage sensor that measures a DC-bus voltage associated with said the grid-integration PV system, a DC-bus voltage control unit that regulates DC-bus voltage and forms a duty-cycle control signal, and a PWM generation unit that generates PWMs for the DC-DC converter from the duty-cycle control signal.

(119) The control system associated with the DC-AC converter control is configured for controlling the DC-AC converter in the grid-integration PV system. The control system can comprise a plurality of sensors that measure a PV output voltage and an AC grid voltage, that outputs the

current of the DC-AC converter. A P&Q calculation unit calculates the real power and the reactive power, and a maximum power point tracking (MPPT) unit maximizes the power acquisition of the PV source in grid-connected mode. A real power control unit regulates real power and forms a control signal according to the derivative of power angle in grid-connected mode. A PV voltage regulation unit builds the linkage between the MPPT unit and the real power control unit, and a reactive power control unit regulates reactive power and forms a control signal of the derivative of output voltage. A P/ω droop unit provides frequency regulation for the AC grid and real power sharing in islanded mode and a Q/E droop unit provides voltage regulation for AC grid and reactive power sharing in islanded mode. The mode detection unit detects the operation mode of the AC grid and generates three digital switch signals. The three digital switches switch the mode of grid-integration PV systems. A voltage-forming unit forms a sinusoidal control voltage according to both the derivative of power angle and the derivative of output voltage passing through the integration units and combining with the global settings. Finally, the PWMs generation unit generates PWMs for the DC-AC converter from the sinusoidal control voltage signal.
(120) In certain embodiments, the DC-DC converter control and the DC-AC converter control are decoupled. Thus, the DC-DC converter control is not affected by mode change in the DC-AC converter control. This can facilitate the seamless mode change in the DC-AC converter control.
(121) The unified control system naturally provides the DC-bus voltage protection, even in the fault conditions of the DC/AC converter or AC grid side. Thus, the fault ride-through performance of the system is enhanced by DC-bus voltage protection.
(122) It should be understood that in certain embodiments, the mode detection unit can be replaced by the external trigger signals, e.g., from a microgrid centralized controller (MGCC). The mode change can be triggered by the mode detection unit or can be triggered by the external signals.
(123) Thus, the unified control framework disclosed herein preserves the continuity of control signals in all control units and the unified control framework has a simple structure and procedure for seamless mode change.
(124) Based on the foregoing, it can be appreciated that a number of embodiments, preferred and alternative, are disclosed herein. In an embodiment a control system, comprises a PV system and a unified control framework that provides mode change between a grid-connected mode and an islanded mode.
(125) In an embodiment, the PV system further comprises a PV collector, a DC-DC converter, and a DC-AC converter with an AC grid integration. In an embodiment the unified control framework further comprises a DC-DC converter control and a DC-AC converter control.
(126) In an embodiment the DC-DC converter control further comprises a voltage sensor that measures a DC-bus voltage associated with said PV system, a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal, and a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal.
(127) In an embodiment the DC-AC converter control further comprises at least one sensor that measures a PV output voltage, at least one sensor that measures an AC grid voltage, and at least one sensor that measures an output current of a DC-AC converter. In an embodiment the DC-AC converter control further comprises a maximum power point tracking (MPPT) unit that maximizes a power acquisition of said PV system in grid-connected mode, a real power control unit that regulates a real power and forms a control signal from a derivative of a power angle in grid-connected mode. a PV voltage regulation unit that provides a linkage between said MPPT unit and said real power control unit, and a reactive power control unit that regulates a reactive power and forms a control signal from a derivative of output voltage. In an embodiment the DC-AC converter control further comprises a P&Q calculation unit that calculates said real power and said reactive power.
(128) In another embodiment the DC-AC converter control further comprises a P/ω droop unit that provides a frequency regulation for an AC grid and real power sharing in islanded mode and a Q/E

droop unit that provides a voltage regulation for said AC grid and reactive power sharing in islanded mode. In an embodiment the DC-AC converter control further comprises a mode detection unit that detects an operation mode of an AC grid, three digital switch signals generated by said mode detection unit, and three digital switches configured to switch the mode of said PV system. In an embodiment the DC-AC converter control further comprises a voltage-forming unit that forms a sinusoidal control voltage according to a derivative of power angle and a derivative of output voltage.

(129) In another embodiment, the DC-DC converter control and said DC-AC converter control are decoupled. In an embodiment the unified control framework is configured for natural DC-bus voltage protection, and a fault ride-through performance of said PV system is enhanced by DC-bus voltage protection.

(130) In an embodiment, the mode change is implemented in said DC-AC converter control. In an embodiment, the unified control framework preserves the continuity of control signals in all control units with the configurations of digital switches.

(131) In another embodiment a system, comprises: a DC-DC converter control comprising: a voltage sensor configured to measure a DC-bus voltage associated with an external system, a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal; a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal; and a DC-AC converter control comprising at least one sensor that measures an output voltage, at least one sensor that measures an AC grid voltage, and at least one sensor that measures an output current of a DC-AC converter.

(132) In an embodiment the DC-AC converter control further comprises a maximum power point tracking (MPPT) unit that maximizes a power acquisition in grid-connected mode, a real power control unit that regulates a real power and forms a control signal from a derivative of a power angle in grid-connected mode a PV voltage regulation unit that provides a link, age between said MPPT unit and said real power control unit, and a reactive power control unit that regulates a reactive power and forms a control signal from a derivative of output voltage.

(133) In another embodiment the DC-AC converter control further comprises a P&Q calculation unit that calculates said real power and said reactive power, a P/ω droop unit that provides a frequency regulation for an AC grid and real power sharing in islanded mode, and a Q/E droop unit that provides a voltage regulation for said AC grid and reactive power sharing in islanded mode. In an embodiment, the DC-AC converter control further comprises three digital switch signals generated by a mode detection unit and three digital switches configured to switch the mode of said PV system.

(134) In another embodiment a control apparatus, comprises a DC-DC converter, a DC-AC converter with an AC grid integration, a voltage sensor that measures a DC-bus voltage associated with said PV system, a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal, a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal, at least one sensor that measures a PV output voltage, at least one sensor that measures an AC grid voltage, and at least one sensor that measures an output current of a DC-AC converter. In an embodiment, the mode change is implemented in said DC-AC converter control, and preserves the continuity of control signals in all control units with the configurations of digital switches.

(135) It will be appreciated that variations of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Also, it should be understood that various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

## Claims

1. A control system, comprising: a PV system; a unified control framework that provides mode change between a grid-connected mode and an islanded mode; a DC-DC converter control; and a DC-AC converter control comprising a voltage-forming unit that forms a sinusoidal control voltage according to a derivative of power angle and a derivative of output voltage.
2. The control system of claim 1 wherein said PV system further comprises: a PV collector; a DC-DC converter; and a DC-AC converter with an AC grid integration.
3. The control system of claim 1 wherein said DC-DC converter control further comprises: a voltage sensor that measures a DC-bus voltage associated with said PV system; a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal; and a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal.
4. The control system of claim 1 wherein said DC-AC converter control further comprises: at least one sensor that measures a PV output voltage; at least one sensor that measures an AC grid voltage; and at least one sensor that measures an output current of a DC-AC converter.
5. The control system of claim 1 wherein said DC-AC converter control further comprises: a maximum power point tracking (MPPT) unit that maximizes a power acquisition of said PV system in grid-connected mode; a real power control unit that regulates a real power and forms a control signal from a derivative of a power angle in grid-connected mode; a PV voltage regulation unit that provides a linkage between said MPPT unit and said real power control unit; and a reactive power control unit that regulates a reactive power and forms a control signal from a derivative of output voltage.
6. The control system of claim 5 wherein said DC-AC converter control further comprises: a P&Q calculation unit that calculates said real power and said reactive power.
7. The control system of claim 1 wherein said DC-AC converter control further comprises: a P/ω droop unit that provides a frequency regulation for an AC grid and real power sharing in islanded mode; and a Q/E droop unit that provides a voltage regulation for said AC grid and reactive power sharing in islanded mode.
8. The control system of claim 1 wherein said DC-AC converter control further comprises: a mode detection unit that detects an operation mode of an AC grid; three digital switch signals generated by said mode detection unit; and three digital switches configured to switch the mode of said PV system.
9. The control system of claim 8 wherein said the unified control framework preserves continuity of control signals in all control units with the digital switches.
10. The control system of claim 1 wherein said DC-DC converter control and said DC-AC converter control are decoupled.
11. The control system of claim 1 wherein said unified control framework is configured for natural DC-bus voltage protection, and a fault ride-through performance of said PV system is enhanced by DC-bus voltage protection.
12. The control system of claim 1 wherein said mode change is implemented in said DC-AC converter control.
13. A system, comprising: a DC-DC converter control comprising: a voltage sensor configured to measure a DC-bus voltage associated with an external system; a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal; and a PWM generation unit that generates PWMs for a DC-DC converter from said duty-cycle control signal; and a DC-AC converter control comprising: at least one sensor that measures an output voltage; at least one sensor that measures an AC grid voltage; and at least one sensor that measures an output current of a DC-AC converter.

14. The system of claim 13 wherein said DC-AC converter control further comprises: a maximum power point tracking (MPPT) unit that maximizes a power acquisition in grid-connected mode; a real power control unit that regulates a real power and forms a control signal from a derivative of a power angle in grid-connected mode; a PV voltage regulation unit that provides a linkage between said MPPT unit and said real power control unit; and a reactive power control unit that regulates a reactive power and forms a control signal from a derivative of output voltage.
15. The system of claim 13 wherein said DC-AC converter control further comprises: a P&Q calculation unit that calculates a real power and a reactive power; a P/ω droop unit that provides a frequency regulation for an AC grid and real power sharing in islanded mode; and a Q/E droop unit that provides a voltage regulation for said AC grid and reactive power sharing in islanded mode.
16. The system of claim 13 wherein said DC-AC converter control further comprises: three digital switch signals generated by a mode detection unit; and three digital switches configured to switch the mode of said PV system.
17. A control apparatus, comprising: a DC-DC converter; a DC-AC converter with an AC grid integration; a voltage sensor that measures a DC-bus voltage associated with a PV system; a DC-bus voltage control unit that regulates said DC-bus voltage and forms a duty-cycle control signal; a PWM generation unit that generates PWMs for said DC-DC converter from said duty-cycle control signal; at least one sensor that measures a PV output voltage; at least one sensor that measures an AC grid voltage; and at least one sensor that measures an output current of a DC-AC converter, wherein the mode change is implemented in a DC-AC converter control, and preserves continuity of control signals in all control units with the configurations of digital switches.